# Alexander Wang

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## EXPERIENCE

**SpaceX (Space Exploration Technologies)** | *Python, Bazel, Structured Text, EtherCAT, Linux* May 2025 – August 2025
*Software Engineering Intern – Satellite Bus Engineering* *Redmond, WA*

- Designed a Beckhoff Programmable Logic Controller (PLC)-based controller architecture to improve safety / reliability for Hardware-in-the-Loop test systems – integrated using Ethernet, ADC / DAC, UART and **reducing downtime by 87%**.
- Created MQTT / ADS protocol drivers for communication between controller and Bazel automated test framework.
- Developed a spacecraft antenna acquisition simulator that models satellite orbital mechanics, multi-axis gimbal kinematics, and RF antenna pattern physics for testing hill-climbing algorithms.
- Designed and built a testbed from scratch using PLCs, PDUs, power supplies, RF attenuators, and networking hardware.

**General Motors** | *MATLAB, Simulink, C++, Python* May 2024 – April 2025
*Software and Controls Intern – Electric Vehicle Propulsion and Thermal Management* *Markham, ON*

- Utilized embedded C to develop thermal control system software for battery, power electronics, and cabin comfort.
- Developed a Software-in-the-Loop testing pipeline – automating performance analysis from **4-5 days to minutes**.

**SAE AutoDrive – Autonomous Vehicle Team (aUToronto)** | *C++, Python, ROS2, Linux* Sept 2023 – June 2025
*State Estimation Lead* *Toronto, ON*

- Led autonomous vehicle team to win **back-to-back 1ST place** finishes at the R2Y3 and R2Y4 SAE AutoDrive Challenge.
- Developed C++ multi-sensor fusion algorithms (Extended Kalman Filter, vision-based lane localization, inertial LiDAR odometry) for state estimation and integrated sensors into pipeline (GPS, IMU, Wheel Encoders, LiDARs, Cameras).

## RESEARCH & PUBLICATIONS

**University of Toronto – Toronto Robotics + AI Lab (TRAIL)** | *Python, AWS, OpenStreetMap* May 2024 – Present
*3D Lane Detection / Labeling for Autonomous Vehicles* *Toronto, ON*

- Developing a Bayesian Attention-based 3D lane detection model and the development of BoreasLane, the first 3D winter condition lane dataset; targeting submission to the Winter Conference on Applications of Computer Vision (WACV).
- Integrated GPS, Camera, and LiDAR sensor data into automatic lane labeling and refinement pipeline.

*Spatial / Temporal Intelligence for Vision–Language–Action Models*

- Developing a hierarchical 4D scene graph framework for Vision–Language–Action models in dynamic environments.

## PROJECTS

**TARS-AI – Open Source Community** | *Python, Raspberry Pi, Fusion360, GitHub* Dec. 2024 – Present
*Co-Founder* *Global*

- Co-founded and led the development of TARS-AI, an open-source robotics community dedicated to creating the robot TARS from the film Interstellar – growing to **1500+ members** and presenting at Open Sauce to **500k+ online fans**.
- Designed a modular software architecture for speech, personality, memory, intent classification, vision and servo control.
- Configured sensors, servos, and peripherals to GPIO, I2C, SPI, and I2S interfaces – troubleshooting with a multimeter.

## EDUCATION

**University of Toronto:** cGPA 3.6/4.0 Toronto, ON
*Engineering Science - Robotics Engineering (AI Minor)* *Expected: April 2026*

**Certificates**: TensorFlow Developer Certificate || Oracle Database SQL Certified Associate
**Courses**: Computer Vision for Robotics || Intro to Learning From Data || Robot Modeling and Control || Mathematics for Robotics

## SKILLS

**Skills**: C/C++, Python, MATLAB, Simulink, Fusion360, Linux, Shell/Bash, AWS, JavaScript, SQL, OracleDB, Git, Jenkins, Bazel, JIRA
**Embedded Systems**: CAN, Ethernet, EtherCAT, LIN, GPIO, I2C, SPI, I2S, UART, ROS/ROS2, Raspberry Pi, STM32, Arduino
**Artificial Intelligence**: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, Hugging Face, numpy, scipy, pandas